

## WORKSHOP ON DEEP LEARNING SESSION

The following article describes recent research on use of AI for supporting the detection of pneumonia:

["UCSD Health Using AI to Identify Pneumonia, Helping COVID-19 Analysis"](#)

In this workshop your aim is to experiment with the use of Data Augmentation to develop a CNN. Before starting this task, you should make sure you understand the example code for the MNIST problem.

1. Download the chest x-ray data from the Blackboard site. The data is split into 3 folders 'test', 'train' and 'val'.
2. Have a look at the distribution of the examples, especially the 'val' directory. Do you think this is appropriate? Change the distribution to make it more suitable (though don't change the 'test' directory). You may want to do this by writing code or just manually. Manually might be better to save time! However, if you want to do this by writing python code, you will need to:

Import os, shutil

And use the methods:

os.listdir(name of dir) to get a list of the file names in a directory

os.mkdir(name of dir) to create a directory

os.path.join(dir,fname) to create a new path name

shutil.copyfile(src\_name, dst\_name) to copy a file

3. Now follow or adapt Cats vs Dogs code we did in the lecture and develop a CNN (provided on the module site - Cats\_v\_Dogs\_aug\_drop.ipynb). Experiment with the architecture and parameters to get the best results you can. Use **a small no of epochs** when experimenting given the computational requirements.
4. What further work would be needed to reach the levels reported in the article and what **ethical and safety** considerations are important?